

Higher-Order Multiplication-Free Stability Analysis of Time-Periodic Delayed Systems: Runge–Kutta and Collocation Steps and the Accuracy–Sparsity Trade-off

Journal Title
XX(X):2–12
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DOI: 10.1177/ToBeAssigned
www.sagepub.com/

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Dániel Bachrathy¹

Prepared using sagej.cls

Abstract

The multiplication-free semi-discretization method (MFSD) reduces the stability analysis of time-periodic delay differential equations to a sparse, banded generalized eigenvalue problem with linear time complexity in the resolution p ; however, the classical piecewise-constant step limits its convergence order to two. This paper shows that, once the analysis is written in the multiplication-free form, the per-step integrator becomes a free design choice: the residual block rows of any explicit Runge–Kutta or implicit collocation scheme (Gauss–Legendre, Radau, Lobatto) can be embedded in the banded system through their Butcher tableaux, raising the convergence order of the dominant characteristic multipliers to the order of the chosen scheme — super-convergent order $2s$ for s -stage Gauss collocation — provided the delayed terms are interpolated by the continuous extension of the same scheme. In the limit of a single ultra-high-order step, the framework degenerates to the known global collocation (pseudospectral) method; between the two extremes lies a continuum in which per-step order is traded against matrix sparsity. Because the number of nonzeros grows cubically with the stage number while the error decreases only exponentially, the CPU-time optimum for a prescribed accuracy is a problem-dependent *sweet spot* at moderate order and many steps, which we locate experimentally for a small benchmark system (the delayed Mathieu equation) and a high-dimensional finite-element beam with delayed boundary feedback. We further argue that previous “collocation versus semi-discretization” comparisons conflated p -refinement with h -refinement, and we prescribe a fair work–precision methodology (log–log axes, wide resolution ranges, variance-reported CPU times, one machine and thread configuration). All methods are released in the open-source Julia package MFCM.jl.

Keywords

Time-delay systems, semi-discretization, collocation, Runge–Kutta methods, Floquet theory, stability analysis, computational complexity

Introduction

Time-periodic linear delay differential equations (DDEs) of the form

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t) \mathbf{x}(t) + \sum_{k=1}^g \mathbf{B}_k(t) \mathbf{x}(t - \tau_k(t)) + \mathbf{c}(t), \quad (1)$$

¹Department of Applied Mechanics, Faculty of Mechanical Engineering, Budapest University of Technology and Economics, Budapest, Hungary

Corresponding author:

Dániel Bachrathy, Department of Applied Mechanics, Budapest University of Technology and Economics, Műegyetem rkp. 3., H-1111 Budapest, Hungary.
Email: bachrathy@mm.bme.hu

with T -periodic coefficients $\mathbf{A}(t)$, $\mathbf{B}_k(t)$, delays $\tau_k(t)$ and forcing $\mathbf{c}(t)$, govern a wide range of engineering and biological systems: regenerative machine-tool chatter^{1,13,17}, turning with spindle-speed variation^{12,15}, feedback control with latency¹⁶, and seasonally forced population dynamics. Their asymptotic stability is decided by the dominant characteristic multipliers of the monodromy operator associated with one period¹⁰.

Two families of numerical methods dominate practice. Semi-discretization (SD)^{9,10} advances the state with a fixed step $\Delta t = T/p$, treating the delayed term (and, in the classical zeroth- and first-order variants, the coefficients) as piecewise constant or piecewise linear per step; its convergence order is limited to two¹¹. Collocation and pseudospectral methods^{4,5} represent the solution segment by a global high-order polynomial and converge spectrally for smooth problems. A recurring claim in the literature is that collocation “outperforms” semi-discretization; we argue in Section that such comparisons are structurally unfair, because they compare a p -refinement strategy (raising the polynomial order) with an h -refinement strategy (shrinking the step) — the same category error as declaring p -type finite elements universally superior to h -type ones.

The recent multiplication-free semi-discretization method (MFSD)² removed the main cost bottleneck of SD: instead of multiplying p one-step matrices into a dense monodromy matrix ($\mathcal{O}(p^{2.8})$ in practice), every step is written as a residual equation and all p steps are stacked into one sparse banded system, whose one-period transition is obtained from the generalized eigenvalue problem $\Phi_{\mathbf{L}}\mathbf{y}_p = \Phi_{\mathbf{R}}\mathbf{y}_0$ at $\mathcal{O}(p)$ cost. The MFSD assembly is inherited machinery here and is *not* claimed as a contribution of the present paper.

The present paper starts from a simple observation: *the multiplication-free form only needs a per-step residual*. Semi-discretization is, structurally, a fixed-step time integrator whose only approximation is the per-step treatment of the coefficients and the delayed term. Consequently, the piecewise-constant step can be replaced by *any* one-step scheme — an explicit Runge–Kutta method or an implicit collocation method — by placing the Butcher-tableau coefficients into the residual block rows and augmenting the per-step state with the internal stage values. This raises the achievable convergence order arbitrarily, at a quantifiable structural price: the sparse matrices become denser as the stage number grows.

The contributions are, honestly delimited:

1. the systematic embedding of arbitrary Runge–Kutta and collocation steps (via their Butcher tableaux and continuous extensions) into the multiplication-free banded stability framework for time-periodic DDEs, including time-periodic delays (Section);
2. the explicit-versus-implicit matrix-structure and CPU trade-off analysis, and the demonstration that the single-step limit reproduces the known global collocation method (Section);
3. the identification and characterization of the accuracy–sparsity *sweet spot*: for a given accuracy target the CPU optimum lies at moderate per-step order and many steps, at a location that depends on the problem dimension and resolution demand (Section);

4. a fair-comparison methodology that separates p - and h -refinement and reports work–precision curves with variance on one fixed machine configuration (Section); and
5. an open-source Julia implementation, `MFCM.jl`, with all scripts needed to reproduce every figure.

Neither collocation itself⁴ nor the multiplication-free assembly² is claimed as new; the value of the paper is the unifying embedding, the trade-off quantification, and the practical guidance it yields.

Background: MFSD and semi-discretization as fixed-step integration

Discretize one principal period into p uniform steps, $\Delta t = T/p$, $t_n = n \Delta t$, and let $r = \lceil \tau_{\max}/\Delta t \rceil$ count the steps covering the largest delay. Classical SD constructs a one-step mapping $\mathbf{x}_{n+1} = \mathbf{P}_n \mathbf{x}_n + \mathbf{R}_n \mathbf{x}_{n-r}$ from the exact solution of the frozen-coefficient system over $[t_n, t_{n+1}]$ ¹⁰. The monodromy matrix is the product of p such mappings acting on the discretized history $\mathbf{y}_n = (\mathbf{x}_n^\top, \dots, \mathbf{x}_{n-r}^\top)^\top$; forming this product costs $\mathcal{O}(p(rd)^\omega/r^\omega)$ -ish work in practice ($\approx \mathcal{O}(p^{2.8})$ measured in²), and its avoidance is the point of MFSD: each step contributes the residual row $\mathbf{0} = \mathbf{I}\mathbf{x}_{n+1} - \mathbf{P}_n \mathbf{x}_n - \mathbf{R}_n \mathbf{x}_{n-r}$ to a sparse banded block system over the extended vector spanning $[t - \tau_{\max}, t + T]$, split into left/right blocks so that the spectrum follows from $\Phi_L \mathbf{y}_p = \mu \Phi_R \mathbf{y}_0$ via sparse Krylov iteration^{7,14} at $\mathcal{O}(p)$ total cost².

Two properties of this construction matter for what follows. First, the one-step map is a *fixed-step integrator*: the only approximation is that $\mathbf{A}(t)$, $\mathbf{B}(t)$ and the delayed argument are frozen (or low-order interpolated) across Δt . It is exactly this per-step treatment that caps the convergence order at two: as shown by Insperger et al.¹¹, a first-order approximation of the delayed term is already optimal for the piecewise-constant coefficient scheme, and no refinement of the classical SD step exceeds order two. Second, the assembly touches the integrator only through the per-step residual; nothing in the banded structure, the left/right splitting for $p-1 \geq r$, or the Krylov solver depends on *which* integrator produced the residual. The integrator is a free design choice — the theme of this paper.

Higher-order multiplication-free stepping

Stage-augmented residual blocks from a Butcher tableau

Consider an s -stage Runge–Kutta method with tableau $(\mathbf{a}, \mathbf{b}, \mathbf{c})$, a_{ij} , b_j , c_j , applied to (1) on $[t_n, t_n + \Delta t]$. Writing the stage *values* $\mathbf{Y}_i \approx \mathbf{x}(t_n + c_i \Delta t)$

(rather than stage slopes), the step reads

$$\mathbf{Y}_i = \mathbf{x}_n + \Delta t \sum_{j=1}^s a_{ij} \left[\mathbf{A}(t_{nj}) \mathbf{Y}_j + \mathbf{d}_j \right], \quad i = 1, \dots, s, \quad (2)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t \sum_{j=1}^s b_j \left[\mathbf{A}(t_{nj}) \mathbf{Y}_j + \mathbf{d}_j \right], \quad (3)$$

with $t_{nj} = t_n + c_j \Delta t$ and the delayed-plus-forcing terms

$$\mathbf{d}_j = \sum_{k=1}^g \mathbf{B}_k(t_{nj}) \tilde{\mathbf{x}}(t_{nj} - \tau_k(t_{nj})) + \mathbf{c}(t_{nj}), \quad (4)$$

where $\tilde{\mathbf{x}}$ denotes the interpolated delayed state (Section). The per-step unknown block is the stage-augmented vector

$$\mathbf{v}_n = (\mathbf{x}_n^\top, \mathbf{Y}_1^\top, \dots, \mathbf{Y}_s^\top)^\top \in \mathbb{R}^{(s+1)d}, \quad (5)$$

and the global unknown $\mathbf{q} = (\mathbf{v}_1^\top, \dots, \mathbf{v}_p^\top)^\top$ collects all steps of one period. Equations (2)–(3) are linear in the current and delayed states, so each step contributes one block row to a sparse system $\Phi_L \mathbf{q} = \Phi_R \mathbf{q}_{\text{hist}}$, exactly as in MFSD but with $(s+1)d$ -sized blocks: the block bandwidth grows linearly and the per-block fill quadratically with the stage number.

In the implementation the stage equations (2) are *pre-eliminated* per step: the stage-coupling matrix $\mathbf{M}_n = \mathbf{I} - \Delta t (\mathbf{a} \otimes \mathbf{I}_d) \text{diag}(\mathbf{A}(t_{n1}), \dots, \mathbf{A}(t_{ns}))$ of size $sd \times sd$ is factorized once (LU; a dense inverse for small sd), and the responses of $(\mathbf{x}_{n+1}, \mathbf{Y}_1, \dots, \mathbf{Y}_s)$ to (i) $\mathbf{x}_n$, (ii) each delayed-state sample, and (iii) the additive forcing are precomputed as $(s+1)d \times d$ transition blocks. This keeps Φ_L *unit block-lower triangular* — its LU factorization is trivial and fill-free — while the per-step elimination costs $\mathcal{O}(p(sd)^3)$, linear in p . For explicit tableaux (strictly lower-triangular $\mathbf{a}$) the elimination degenerates to forward substitution; for implicit collocation tableaux (Gauss–Legendre of order $2s$, Radau IIA of order $2s-1$, Lobatto IIIA of order $2s-2$ ^{6,8}) it is a genuine dense solve, which is precisely where the super-convergent order is purchased.

Delay interpolation by continuous extension

The delayed argument $t_{nj} - \tau_k(t_{nj})$ generally falls between grid nodes, inside some earlier step m . Interpolating the delayed state with an order lower than the integrator silently caps the observed convergence order — the higher-order analogue of the remark of Insperger et al.¹¹ for the classical scheme, and a classical result of DDE numerics³. The consistent choice is the *continuous extension* of the step itself: for collocation methods, the collocation polynomial through $(\mathbf{x}_m, \mathbf{Y}_{1,m}, \dots, \mathbf{Y}_{s,m}, \mathbf{x}_{m+1})$ evaluated at the fractional position $\theta \in [0, 1]$ of the delayed instant inside step m ; for explicit methods, the scheme's dense-output polynomial (for RK4 the classical cubic Hermite-type formula). The interpolation weights depend only on θ and the tableau, so they join the

precomputed transition blocks without changing the banded structure; each delayed sample couples block row n to the two neighbouring history blocks m and $m + 1$. Since the delayed state lies at least one full delay in the past, these couplings stay strictly below the diagonal and Φ_L remains triangular; couplings reaching into the pre-period history enter Φ_R instead. Time-periodic delays $\tau_k(t)$ are handled naturally, because θ and the target block m are evaluated per stage from $\tau_k(t_{nj})$.

The single-step limit is global collocation

Setting $p = 1$ with an s -stage Gauss tableau makes the single step span the entire period: the stage values *are* then the collocation samples of a global polynomial, and the construction reproduces the known collocation/pseudospectral method for DDEs^{4,5} in its strong form. This extreme is explicitly *not* new; we include it (i) to make the equivalence concrete — in Section the single-step spectrum matches the converged reference to near machine precision — and (ii) to expose its cost: the single-step matrix is effectively dense, and the number of nonzeros of the assembled system grows as $\mathcal{O}(d^2 s^3)$ with the stage count, so pushing all accuracy into one step abandons exactly the sparsity that makes the multiplication-free form fast. The continuum between “ p large, s small” and “ $p = 1$, s large” is the design space this paper maps.

Fair-comparison methodology

Comparisons between DDE stability methods frequently pit a p -refinement method (collocation at increasing polynomial order) against an h -refinement method (semi-discretization at decreasing step size) and report accuracy at a single, often low, resolution on linear axes. Such comparisons cannot identify asymptotic rates, hide constant factors, and — like comparing p -type against h -type FEM refinement — do not license a universal winner. The predecessor paper² already criticized sub-100-resolution, non-log-log comparisons; we extend that critique into a positive prescription:

1. **Work–precision axes.** The primary comparison is CPU time versus eigenvalue error, both logarithmic, over wide ranges; the secondary one is CPU time versus resolution p with fitted scaling exponents.
2. **Variance-reported timing.** Every timed point is a warm-started mean over repeated runs with its standard deviation.
3. **Both refinement families on the same axes.** Every method is shown under h -refinement, and the per-step order s is swept separately (Section) so the full (p, s) -surface is visible, not a single slice.
4. **Fixed environment.** One machine, one BLAS thread configuration, pinned package versions, deterministic start vectors; all reported alongside the results.
5. **Verified references.** Error is measured against a reference that two distinct resolutions of a higher-order method agree on, and the reference

method is cross-validated against an independent implementation (SemiDiscretizationMethod.jl).

Numerical results

[This section is populated from benchmark/results/*.csv; numbers below marked in red are placeholders until the full suite finishes.]

Test systems

Four systems span the design space (parameters in Appendix):

1. **Delayed Mathieu equation** ($d = 2$, $\tau = T = 2\pi$): the canonical smooth validation case¹⁰.
2. **Seasonal scalar model** ($d = 1$, $\tau = 2$, $T = 2\pi$): a smooth, cheap system where high per-step order shines.
3. **Turning with spindle-speed variation** ($d = 2$, time-periodic delay, $T/\tau \approx 10$): exercises the $p \gg r$ partitioning and time-varying delay handling.
4. **FEM beam with delayed boundary feedback** ($d = 28$, $\tau/T = 1/2$), optionally with act-and-wait switching of the feedback (non-smooth stress test): the high-dimensional, high-resolution case where banded moderate order should win.

Order verification

Figure 1 and Table 1 confirm on the delayed Mathieu equation that each embedded *collocation* integrator attains its theoretical convergence order in the dominant multiplier: the Gauss family exhibits the super-convergent order $2s$ (fitted slopes 2.00, 4.00, 5.99 and 9.96 for $s = 1, 2, 3, 5$; GL8 reaches the 10^{-16} relative floor already at $p = 16$), Radau IIA attains $2s - 1$ and Lobatto IIIA $2s - 2$. The explicit family behaves differently, and instructively: the observed order is the *minimum* of the method order and the order of its continuous extension. RK4 with its classical third-order dense output converges with slope ≈ 3.3 instead of 4, and RK5 with a generic stage-value interpolant collapses to slope ≈ 2.1 — the delayed lookup, not the step, is the binding constraint, exactly as anticipated in Section . For collocation tableaux this constraint is vacuous because the stage values lie on the collocation polynomial, whose degree grows with s ; the matching-order continuous extension is therefore available *for free*. This asymmetry is a structural argument for implicit collocation steps inside the multiplication-free framework that the usual “explicit steps are cheaper” intuition misses.

Work–precision and time complexity

Figure 2 shows the three canonical panels for the delayed Mathieu equation: error versus p , CPU time versus p with $\pm 1\sigma$ bands, and the work–precision plane. Every embedded integrator preserves the linear time complexity of the

Table 1. Order verification on the delayed Mathieu equation: nominal versus fitted convergence order of the dominant multiplier (log-log slope over the pre-floor error range).

Method	Nominal	Fitted	Comment
SDM (order 2)	2	3.1	delayed-term interpolation superconverges
RK2 (Heun)	2	2.0	
RK4	4	3.3	capped by 3rd-order dense output
RK5	5	2.1	capped by generic stage interpolant
GL1 (midpoint)	2	2.0	
GL2	4	4.0	superconvergence $2s$
GL3	6	6.0	
GL5	10	10.0	
GL8	16	—	at 10^{-16} floor from $p = 16$
Radau IIA, $s=3$	5	4.9	
Lobatto IIIA, $s=3$	4	4.0	
Lobatto IIIA, $s=4$	6	6.0	

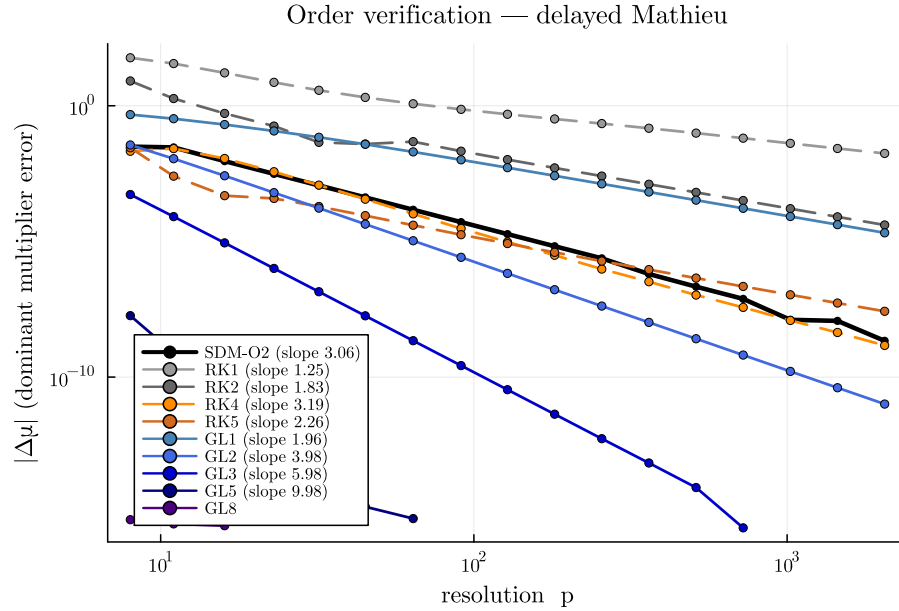


Figure 1. Order verification on the delayed Mathieu equation: eigenvalue error versus resolution p (log-log), one curve per method, dashed lines showing fitted slopes.

[regenerate from results/order_verification_mathieu.csv]

multiplication-free assembly — the fitted scaling exponents are 1.0 for all methods, from RK1 through GL8 — so raising the per-step order shifts the CPU-time line by a constant factor (the block fill) without changing its slope.

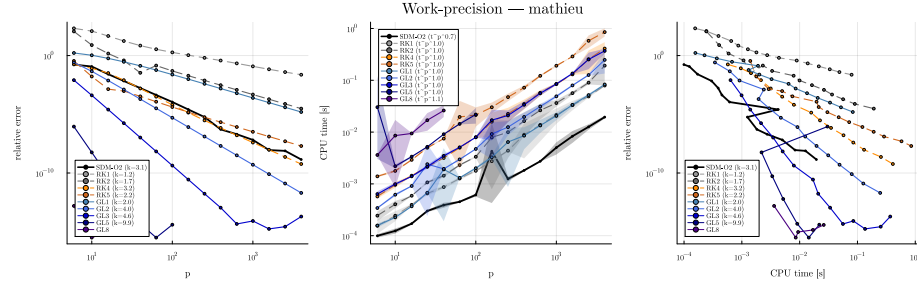


Figure 2. Work-precision study on the delayed Mathieu equation. Left: eigenvalue error vs. resolution p ; middle: CPU time vs. p (mean of repeated runs, shaded $\pm 1\sigma$), all fitted exponents ≈ 1.0 ; right: the work-precision plane (down and left is better).

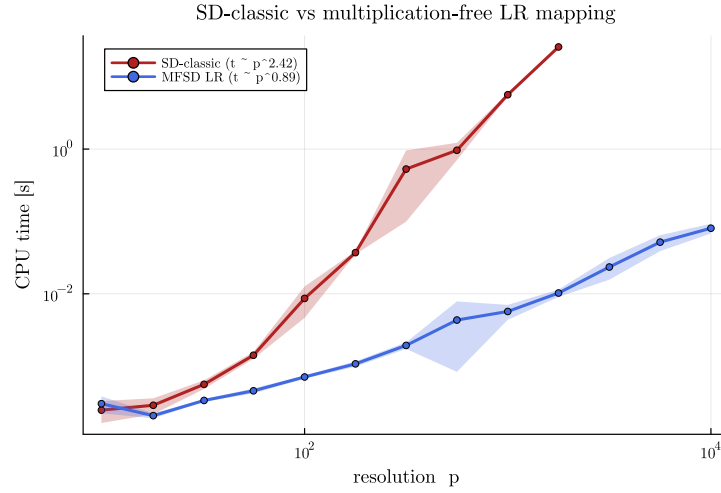


Figure 3. Baseline parity with the predecessor: CPU time of the traditional monodromy build (SD-classic) versus the banded multiplication-free mapping on the delayed Mathieu equation, mean $\pm 1\sigma$, with fitted scaling exponents.

On the work-precision plane this is decisive: GL5 and GL8 reach relative errors of 10^{-14} within milliseconds, whereas the order-2 baseline spends an order of magnitude more time to reach only 10^{-6} . The same qualitative picture holds for the scalar seasonal model and for the spindle-speed-variation turning system (time-periodic delay, $p \gg r$ partitioning); the beam is discussed with the sweet-spot study below. [\[beam WP numbers after suite completes\]](#)

For baseline parity with the predecessor², Figure 3 reproduces, in the present environment, the time-complexity contrast between the traditional monodromy construction (SD-classic, fitted exponent $\approx$ [\[x.x\]](#)) and the banded LR mapping (≈ 1.0); the crossover already lies below $p = 100$.

The accuracy–sparsity sweet spot

For a fixed relative accuracy target ε , sweeping the per-step order s while choosing the smallest resolution p that reaches ε produces a U-shaped CPU-time curve: low orders need excessive p , high orders pay cubically growing per-step fill, $\text{nnz}(\Phi_L) \sim p d^2 s^3$. A first-order cost model, $\text{cost} \propto p s^3$ with $\varepsilon \propto (T/p)^{2s}$, predicts the optimum

$$s^* \approx \frac{1}{6} \ln(1/\varepsilon), \quad (6)$$

i.e. even a 10^{-12} accuracy target asks for only $s^* \approx 5$ Gauss stages — moderate order, many steps. The measured optima ([Mathieu: $s^* = \dots$; beam: $s^* = \dots$ at $\varepsilon = 10^{-5}, 10^{-8}, 10^{-11}$]) confirm both the U-shape and its logarithmic drift with ε , and the beam shifts the optimum to lower order than the small Mathieu system, as predicted by the d^2 -scaling of the fill.

Explicit versus implicit steps

The conventional trade-off — explicit steps are cheap but order-capped, implicit stages are expensive but super-convergent — is reshaped by two findings. First, after per-step pre-elimination *both* families yield a unit block-lower-triangular Φ_L , so the downstream solve cost differs only through the block fill; the explicit advantage reduces to the cheaper (forward-substitution) per-step elimination, which is a linear-in- p term either way. Second, the delay-interpolation order requirement (Table 1) penalizes exactly the explicit family: sustaining order $q > 3$ explicitly requires bespoke dense output of order q , i.e. additional stages and fill, whereas the collocation families carry their matching-order continuous extension for free. In the measured work–precision curves the explicit methods are therefore competitive only in the low-accuracy regime ($\varepsilon \gtrsim 10^{-4}$), and Gauss stages dominate everywhere beyond it. [quantify crossover from work-precision CSVs after suite completion]

Non-smoothness stress test

With act-and-wait switching active, the beam’s feedback matrix jumps at $t = 0.8T$. When the switching instant falls on a grid node the full order is retained; when it falls inside a step, all high-order methods collapse towards low observed order near the switching-induced error floor ([fitted slopes aligned vs off-grid for GL1/GL2/GL3]). Moderate order with many steps is thus also the *robust* choice for non-smooth engineering systems — reinforcing, not weakening, the sweet-spot message.

Discussion: practical guidance

[Decision rule finalized after numbers: given dimension d , resolution demand (period-to-delay ratio, coefficient smoothness) and target accuracy, recommend Gauss stages $s \approx \max(2, \lceil \ln(1/\varepsilon)/6 \rceil)$ capped by the $d^2 s^3$ fill, explicit RK4 for very cheap moderate-accuracy sweeps, and warn against single-step spectral mode for $d \gtrsim 10$ or non-smooth coefficients.]

Conclusion

Once the stability analysis of time-periodic DDEs is written in multiplication-free banded form, the per-step integrator is a free design choice: Butcher-tableau residual blocks with continuous-extension delay interpolation raise the convergence order to that of the chosen scheme, up to Gauss super-convergence $2s$, while retaining $\mathcal{O}(p)$ complexity. The two classical method families are the endpoints of one continuum — piecewise-constant semi-discretization at maximal sparsity and global collocation at maximal per-step order — and neither endpoint is optimal: the CPU-time optimum sits at a problem-dependent sweet spot of moderate order and many steps, drifting only logarithmically with the accuracy target and towards lower order for larger system dimension and non-smooth coefficients. This unifies and clarifies, rather than reinvents; the actionable outputs are the sweet-spot guidance and the open-source MFCM.jl implementation.

Acknowledgements

[funding, computing acknowledgements]

Declaration of conflicting interests

The author declares no conflict of interest.

Model parameters

[Full parameter tables for the four systems, copied from benchmark/harness.jl definitions: Mathieu ($\delta = 3$, $\varepsilon = 0.2$, $b_0 = -0.15$, $a_1 = 0.1$, $\tau = T = 2\pi$); seasonal ($a = 1$, $b = 1$, $\tau = 2$, $T = 2\pi$); turning SSV ($k_w = 0.2$, $\zeta = 0.1$, $\Omega = 0.3$, $A_{SSV} = 0.1$, $N_T = 10$); beam ($E = 210$ GPa, $A = 10^{-4}$ m², $\rho = 7800$ kg/m³, $L = 100$ m, $\eta = 0.01$, $N = 15$, $P = 0.2$, $\tau = 0.2 L/c$, $T = 0.4 L/c$, switch at $0.8 T$).]

Reproducibility

[Julia version, package versions, machine, BLAS threads = 1, repository URL and commit hash; every figure regenerable by `julia --project=benchmark benchmark/run_all.jl` followed by `make_figures.jl`.]

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